Project Title
GrabSnack Web Application – Google OAuth2 Login Integration

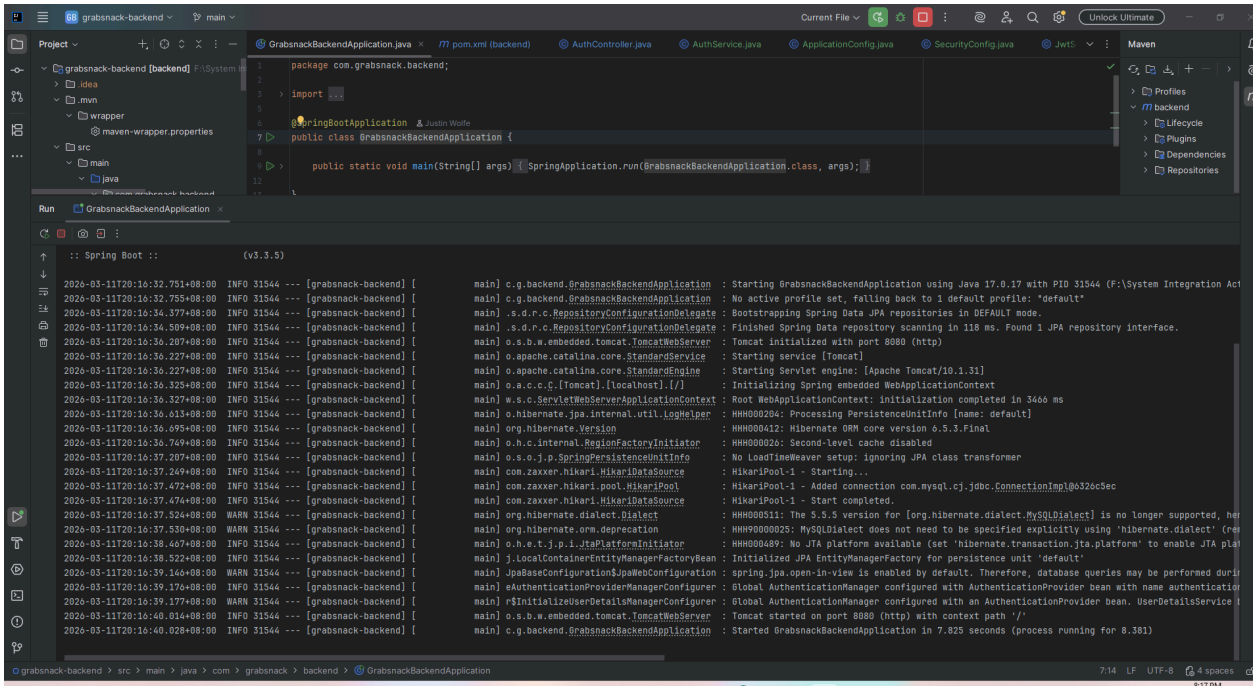
Brief Description of How Google Login Was Implemented
Google Login was implemented using Spring Boot's OAuth2 authentication framework. When a user clicks the "Login with Google" button on the React frontend, the application redirects the user to Google's authentication page. After successful authentication, Google returns the user information to the backend through the OAuth2 callback endpoint. The backend then retrieves the user's email and profile information, checks if the user exists in the database, and creates a new user record if necessary. A JWT token is generated for authenticated users and the system redirects them back to the frontend with the authentication token.

Backend Technologies / Libraries Used
The backend was built using Spring Boot with several supporting libraries. Spring Security was used to handle authentication and authorization, while Spring Security OAuth2 Client was used for Google authentication integration. Spring Data JPA and Hibernate were used for database interaction with MySQL. JWT (JSON Web Token) was implemented to manage secure user sessions after authentication.

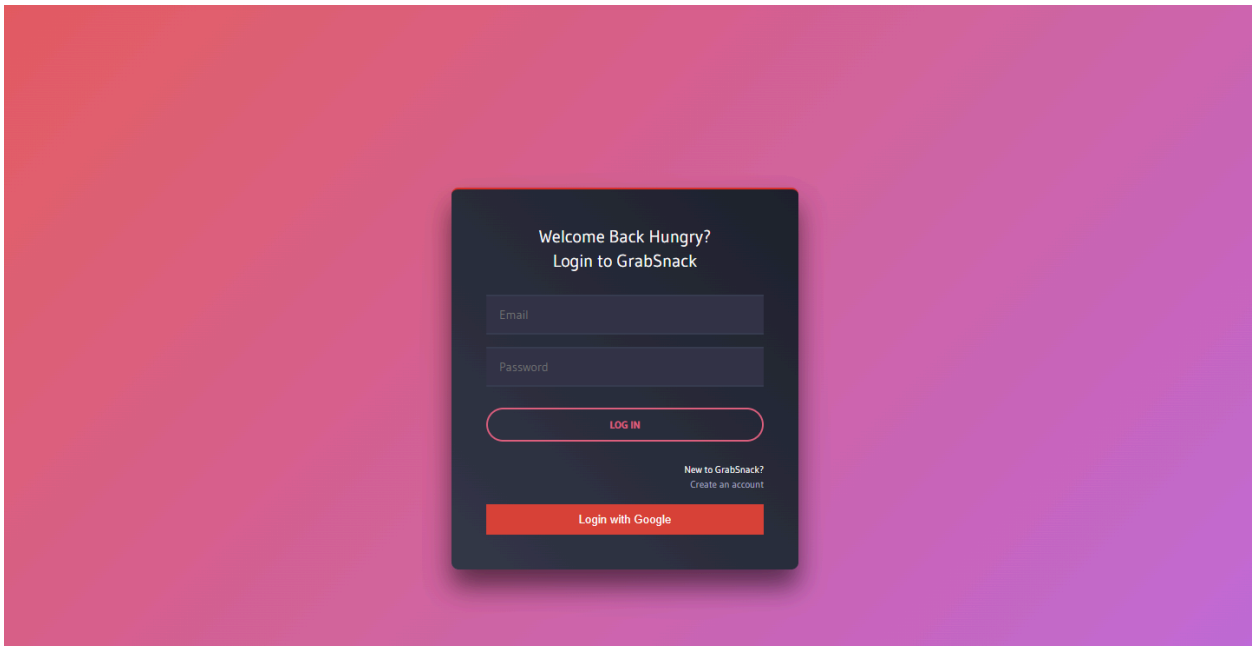
Frontend Integration Summary
The frontend was developed using React.js. A Google Login button was added to the login page that redirects users to the backend OAuth2 authorization endpoint. After successful login, the backend redirects the user back to the frontend with the generated JWT token and user information. The frontend then uses this token for authenticated requests to protected backend APIs.

Challenges Encountered
Several challenges were encountered during the implementation process. These included configuring the correct OAuth2 redirect URI in the Google Cloud Console, resolving dependency issues in the Spring Boot configuration, and handling backend authentication provider setup. Debugging OAuth errors and ensuring proper communication between the React frontend and Spring Boot backend were also key challenges during development.

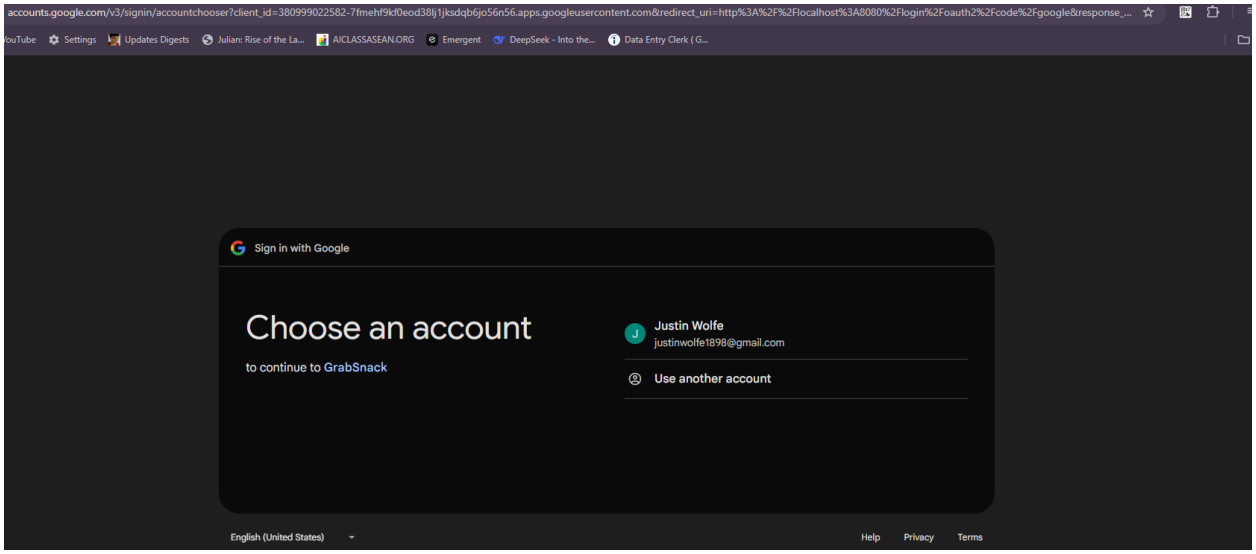
GitHub GrabSnack : https://github.com/Jeepittea/Grabsnack
SpringBoot - Back end


grabsnack-backend
main
Current File
Unlock Ultimate
Project
GrabsnackBackendApplication.java
pom.xml (backend)
AuthController.java
AuthService.java
ApplicationConfig.java
SecurityConfig.java
Maven
grabsnack-backend [backend]
.idea
.mvn
wrapper
maven-wrapper.properties
src
main
java
package com.grabsnack.backend;
import ...
@SpringBootApplication
Justin Wolfe
public class GrabsnackBackendApplication {
public static void main(String[] args) { SpringApplication.run(GrabsnackBackendApplication.class, args); }
Profiles
backend
Lifecycle
Plugins
Dependencies
Repositories
Run
GrabsnackBackendApplication
:: Spring Boot :: (v3.3.5)
2026-03-11T20:16:32.751+08:00 INFO 31544 --- [grabsnack-backend] [main] c.g.backend.GrabsnackBackendApplication : Starting GrabsnackBackendApplication using Java 17.0.17 with PID 31544 (F:\System Integration Act
2026-03-11T20:16:32.755+08:00 INFO 31544 --- [grabsnack-backend] [main] c.g.backend.GrabsnackBackendApplication : No active profile set, falling back to 1 default profile: "default"
2026-03-11T20:16:34.377+08:00 INFO 31544 --- [grabsnack-backend] [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-03-11T20:16:34.509+08:00 INFO 31544 --- [grabsnack-backend] [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 118 ms. Found 1 JPA repository interface.
2026-03-11T20:16:36.207+08:00 INFO 31544 --- [grabsnack-backend] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-03-11T20:16:36.227+08:00 INFO 31544 --- [grabsnack-backend] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-03-11T20:16:36.227+08:00 INFO 31544 --- [grabsnack-backend] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.31]
2026-03-11T20:16:36.325+08:00 INFO 31544 --- [grabsnack-backend] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-03-11T20:16:36.327+08:00 INFO 31544 --- [grabsnack-backend] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 3466 ms
2026-03-11T20:16:36.613+08:00 INFO 31544 --- [grabsnack-backend] [main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-03-11T20:16:36.695+08:00 INFO 31544 --- [grabsnack-backend] [main] org.hibernate.Version : HHH000412: Hibernate ORM core version 6.5.3.Final
2026-03-11T20:16:36.749+08:00 INFO 31544 --- [grabsnack-backend] [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-03-11T20:16:37.207+08:00 INFO 31544 --- [grabsnack-backend] [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-03-11T20:16:37.249+08:00 INFO 31544 --- [grabsnack-backend] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-03-11T20:16:37.472+08:00 INFO 31544 --- [grabsnack-backend] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@6326c5ec
2026-03-11T20:16:37.474+08:00 INFO 31544 --- [grabsnack-backend] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-03-11T20:16:37.524+08:00 WARN 31544 --- [grabsnack-backend] [main] org.hibernate.dialect.Dialect : HHH000511: The 5.5.5 version for [org.hibernate.dialect.MySQLDialect] is no longer supported, her
2026-03-11T20:16:37.530+08:00 WARN 31544 --- [grabsnack-backend] [main] org.hibernate.orm.deprecation : HHH90000025: MySQLDialect does not need to be specified explicitly using 'hibernate.dialect' (re
2026-03-11T20:16:38.467+08:00 INFO 31544 --- [grabsnack-backend] [main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA plat
2026-03-11T20:16:38.522+08:00 INFO 31544 --- [grabsnack-backend] [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-03-11T20:16:39.146+08:00 WARN 31544 --- [grabsnack-backend] [main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed durin
2026-03-11T20:16:39.176+08:00 INFO 31544 --- [grabsnack-backend] [main] eAuthenticationProviderManagerConfigurer : Global AuthenticationManager configured with AuthenticationProvider bean with name authenticatio
2026-03-11T20:16:39.177+08:00 WARN 31544 --- [grabsnack-backend] [main] r$InitializeUserDetailsManagerConfigurer : Global AuthenticationManager configured with an AuthenticationProvider bean. UserDetailsService b
2026-03-11T20:16:40.014+08:00 INFO 31544 --- [grabsnack-backend] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-03-11T20:16:40.028+08:00 INFO 31544 --- [grabsnack-backend] [main] c.g.backend.GrabsnackBackendApplication : Started GrabsnackBackendApplication in 7.825 seconds (process running for 8.381)
grabsnack-backend > src > main > java > com > grabsnack > backend > GrabsnackBackendApplication
7:14
LF
UTF-8
4 spaces

Login


Welcome Back Hungry?
Login to GrabSnack
Email
Password
LOG IN
New to GrabSnack?
Create an account
Login with Google

Login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YouTube
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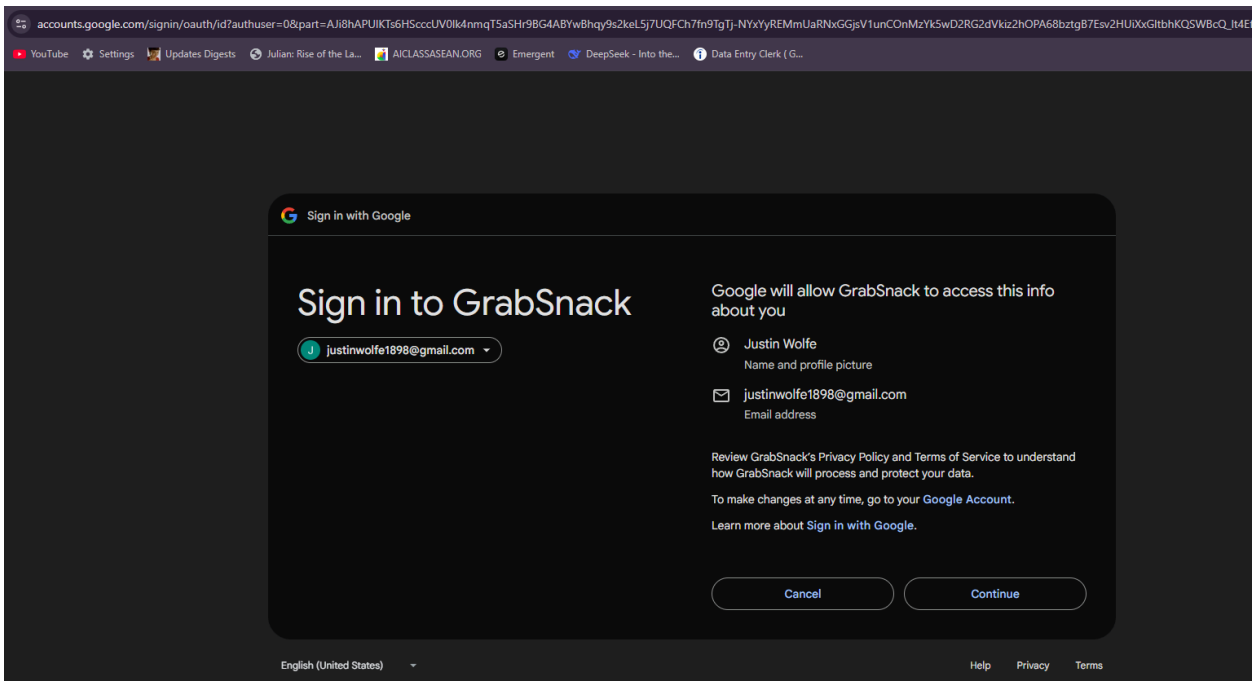
Updates Digests
Julian: Rise of the La...
AICLASSASEAN.ORG
Emergent
DeepSeek - Into the...
Data Entry Clerk (G...
Sign in with Google
Choose an account
to continue to GrabSnack
Justin Wolfe
justinwolfe1898@gmail.com
Use another account
English (United States)
Help
Privacy
Terms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YouTube
Settings
Updates Digests
Julian: Rise of the La...
AICLASSASEAN.ORG
Emergent
DeepSeek - Into the...
Data Entry Clerk (G...
Sign in with Google
Sign in to GrabSnack
justinwolfe1898@gmail.com
Google will allow GrabSnack to access this info about you
Justin Wolfe
Name and profile picture
justinwolfe1898@gmail.com
Email address
Review GrabSnack's Privacy Policy and Terms of Service to understand how GrabSnack will process and protect your data.
To make changes at any time, go to your Google Account.
Learn more about Sign in with Google.
Cancel
Continue
English (United States)
Help
Privacy
Terms